Mailer — User Guide

Mailer is a Niagara Program Object that emails files from the station (reports, CSVs, logs) through the station's EmailService. Each file line can go to its own recipients, and a wildcard line can send just the newest matching file.

Covers **N4 v1.03 / v1.04**.

Contents

1. How it works in one minute
2. Before you start: the EmailService
3. Every field, explained
4. The files list
5. Who gets which email
6. Subject and body placeholders
7. Actions
8. Reading status
9. Complete setups (recipes)
10. Problems and fixes

---

1. How it works in one minute

```
files (one line per file / wildcard / folder, optional "=> recipients")
  |
Send Now ──►  finds the files ──► checks every address ──► one email per recipient group
                                                              └─► EmailService outgoing account
  |
status: SENT / FAILED / DRY RUN  +  one block per email
```

- **Nothing happens on its own.** Press **Send Now**, or link something to the `send` action — e.g. a Trigger Schedule for a monthly email, or another program.
- It checks everything first: if any file line matches nothing, or any address is malformed, **nothing** is sent and `status` says which.
- The built-in `execute` action does nothing on Mailer (you can hide it).

---

2. Before you start

Mailer sends through the station's **EmailService** (Config → Services → EmailService). It needs at least one outgoing account set up there (SMTP server, port, login). Test it from the EmailService first. After adding Mailer, run **Refresh Accounts** once so the `sender` drop-down lists the accounts.

---

3. Every field, explained

Legend: **Set** = you set it · **Out** = the program writes it (read only).

About

Field		What it is
`version`	Out	The program version.
`quickGuide`	Out	Short built-in help.

Last send

Field		What it is
`status`	Out	First line `SENT`, `FAILED` or `DRY RUN`, then one block per email. See Reading status.
`lastSent`	Out	Date and time of the last send that got at least one email out.

Who it comes from

Field		Default	What it is
`sender`	Set	Default	Drop-down of the EmailService's outgoing accounts. **Default** = the first one. Filled in on station start and by **Refresh Accounts**.
`fromAddress`	Set	blank	The From address. Blank = the account's own address. Some mail servers refuse a From that isn't the account's.

What to send

Field		Default	What it is
`files`	Set	blank	One line per file, wildcard or folder; `#` lines are ignored; `=> addresses` on a line sends it to its own recipients. See The files list. Blank = send the body only, to `to`.
`latestOnly`	Set	false	**true** = each wildcard or folder line attaches only its newest file (by modified date). To do it for one line only, start that line with `latest:`.

Who gets it

Field		What it is
`to`	Set	Recipients for file lines **without** `=>` (and for a body-only email).
`cc`	Set	Added to **every** email.
`bcc`	Set	Added to **every** email.

Addresses can be separated by `;`, `,`, spaces or new lines. `Name <a@company.com>` uses `a@company.com`.

The message

Field		Blank =	What it is
`subject`	Set	`Automated Report - {reportname}`	The subject. Placeholders allowed.
`body`	Set	`This is automated report {reportname} autogenerated @ {time} {date}`	The email text. Placeholders allowed. Same on every email.

Saved settings and logs

Field		Blank =	What it is
`configFile`	Set	folder `file:^Mailer_Config`	Where **Save Config / Load Config** keep the settings. A file (`file:^Backups/Reports.properties`) or a folder (`file:^Backups` → `Backups/<program name>.properties`).
`logFilePath`	Set	`file:^logs/Mailer.log`	A new log for each Send / Dry Run; the old one is archived.
`maxArchives`	Set	10	Old logs kept. 0 = keep all.

---

4. The files list

Paths start at the top of the station's Files (`file:^` = the station folder; `file:` is optional).

Line	Attaches
`file:^Reports/Summary.csv`	that file
`file:^Reports/Monthly/*.csv`	every `.csv` in `Reports/Monthly` (`*` and `?` work in the name)
`file:^Reports/Monthly`	every file in that folder (not sub-folders)
`latest: file:^Reports/Monthly/*.csv`	only the newest `.csv` there

Line	Attaches
`# file:^Reports/Old.csv`	nothing — a comment

- A wildcard only looks in the folder you name: `file:^B*.csv` looks in the top folder only. Use `file:^Reports/B*.csv` for files in `Reports`.
- **Newest is per line.** `latest: file:^Reports/B*.csv` gives **one** file in total, not one per building — write one line per building for that.
- Paths outside the station folder are refused (`PATH REFUSED`).

---

5. Who gets which email

Add `=> addresses` to a line to send that line's files to its own people. Prefix with `cc:` or `bcc:`; no prefix = To.

```
file:^Reports/Report BXXXXX.csv => a@company.com; cc: b@company.com
file:^Reports/Report BYYYYY.csv => c@company.com bcc:d@company.com
file:^Reports/Summary.csv
```

- **Each line is one email.** Lines with exactly the same recipients are combined into one email with all their files.
- Lines without `=>` go to `to`.
- `cc` and `bcc` (the slots) are added to every email.
- A line needs a To or Cc. Bcc alone isn't enough; a line with only Cc sends to them as To.
- Subject and body are the same on every email; `{reportname}` changes with each email's files.

---

6. Subject and body placeholders

Placeholder	Becomes
`{reportname}`	The attached file name(s) without the extension, joined with `,`
`{time}`	Send time, `HH:mm` (24 h)
`{date}`	Send date, `yyyy-MM-dd`

Case doesn't matter. Example: `Monthly report {reportname} - {date}` → `Monthly report Report BXXXXX - 2026-11-01`.

---

7. Actions

Action (display name)	What it does
send (Send Now)	Send. Link a trigger here to send automatically.
dryRun (Dry Run - No Send)	Shows each email that would go out — recipients, files, sizes and modified dates. Nothing is sent.
refreshAccounts	Rebuild the `sender` drop-down after adding or removing accounts.
saveConfig	Save files, to, cc, bcc, subject, body, fromAddress, latestOnly and sender to `configFile`. An existing file is first copied to `.bak`.
loadConfig	Load them back. If the saved account no longer exists, Default is picked and `status` says so.

The config file is plain text (`key=value`), editable in Notepad.

---

8. Reading status

```
SENT [2026-11-01 07:00:04] 2 email(s)
Email 1 To: a@company.com  Cc: b@company.com
  Subject: Automated Report - Report BXXXXX
  Reports/Report BXXXXX.csv (12 KB, 2026-11-01 06:55)
  1 file(s), 12 KB - sent
Email 2 To: c@company.com  Bcc: d@company.com
  ...
```

- Line 1 starts with **SENT**, **FAILED** or **DRY RUN**, so another program can check it.
- **FAILED** on line 1 means at least one email failed — the others may have gone out; read the blocks.
- **SENT** means handed to the outgoing account's queue. If it never arrives, look at that account under the EmailService: `numberSent` and `lastDiscardCause`.

When some fail: `FAILED [time] 1 of 2 email(s) failed, 1 sent`, and the failed block ends with `- FAILED: <reason>`.

---

9. Complete setups

Placeholders throughout.

A. One report to one person

Field	Value
files	`file:^Reports/Summary.csv`
to	`a@company.com`

Dry Run → Send Now.

B. Monthly: each building's newest report to its own manager

```
latest: file:^Reports/Monthly VAV health BXXXXX*.csv => a@company.com
latest: file:^Reports/Monthly VAV health BYYYYY*.csv => b@company.com; cc: c@company.com
```

Field	Value
bcc	`you@company.com` (a copy of everything to you)
subject	`Monthly VAV health - {reportname}`

Link a **Trigger Schedule** (set to the 1st of the month, after the reports are made) to the `send` action.

C. Driven by another program

Another program sets `files` / `to` / `subject` / `body`, then invokes `send`, and reads line 1 of `status` for the result.

D. Moving the settings to a new Mailer

On the old one: **Save Config**. On the new one: give it the same name (leave `configFile` blank) or point `configFile` at the same file, then **Load Config**.

---

10. Problems and fixes

status says	Fix
`FAILED … nothing sent` then `not found: <line>`	That line matches no file. Check the path and wildcard; wildcards only look in the named folder.
`bad address: '…' (…)`	Fix that address (one `@`, a dotted domain, no spaces).
`no To or Cc recipient for: <line>`	Give the line `=> addresses`, or fill in `to`.
`no files and no valid address in 'to'`	Nothing to send — fill in `files` or `to`.
`PATH REFUSED …`	The path is outside the station folder.

status says	Fix
`EmailService not found in this station`	Add and set up the EmailService.
`sender '…' not found; available: …`	The account was removed or renamed — **Refresh Accounts** and pick another.
`no config file at …` (Load Config)	Check `configFile`, or run Save Config first.
SENT but nothing arrives	Check the EmailService account (`lastDiscardCause`), spam folders, and whether the server accepts the `fromAddress`.